

FAKE NEWS DETECTION

PREPARED BY:

Eliana Kawira

Daniel Mwaka

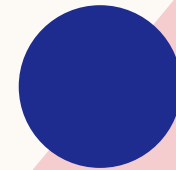
Marion Osong'

Gerald Wanjala

Abdirahman yussuf

CONTENTS

- ☐ Introduction
- ☐ Business problem
- ☐ Objectives
- ☐ Findings
- ☐ Conclusion
- ☐ Recommendations
- ☐ Next steps



INTRODUCTION

BUSINESS UNDERSTANDING

The spread of misinformation and fake news through online news dissemination platforms and social media platforms has become a serious threat to public trust, democracy, health communication, and media integrity.

The ability to automatically distinguish fake from real news is essential for:

- ❑ Social media platforms
- ❑ Fact-checking organizations.
- ❑ News aggregators
- ❑ The general public.



PROBLEM STATEMENT

- The proliferation of fake news on digital platforms poses a significant risk to public trust and decision-making.
- Traditional methods of identifying fake news are slow, and resource-intensive.

PROPOSED SOLUTION

- Develop an automated binary classification system using transformer-based Natural Language Processing (NLP) models to detect and flag fake news articles.

Potential Impact: Avail a scalable solution to help identify harmful misinformation, improve information credibility and support fight against digital manipulation.



OBJECTIVES

1. Perform Exploratory Data Analysis.
2. Build a baseline Logistic Regression model to predict an article's class (fake or real).
3. Build an XGBoost, a LSTM neural network and finetune the RoBERTa-base transformer.
4. Explain the best-fit model's predictions at feature level.
5. Deploy selected model.

DATASET

Datasets used;

- ❖ gossipcop_fake.csv
- ❖ gossipcop_real.csv
- ❖ politifact_fake.csv
- ❖ politifact_real.csv

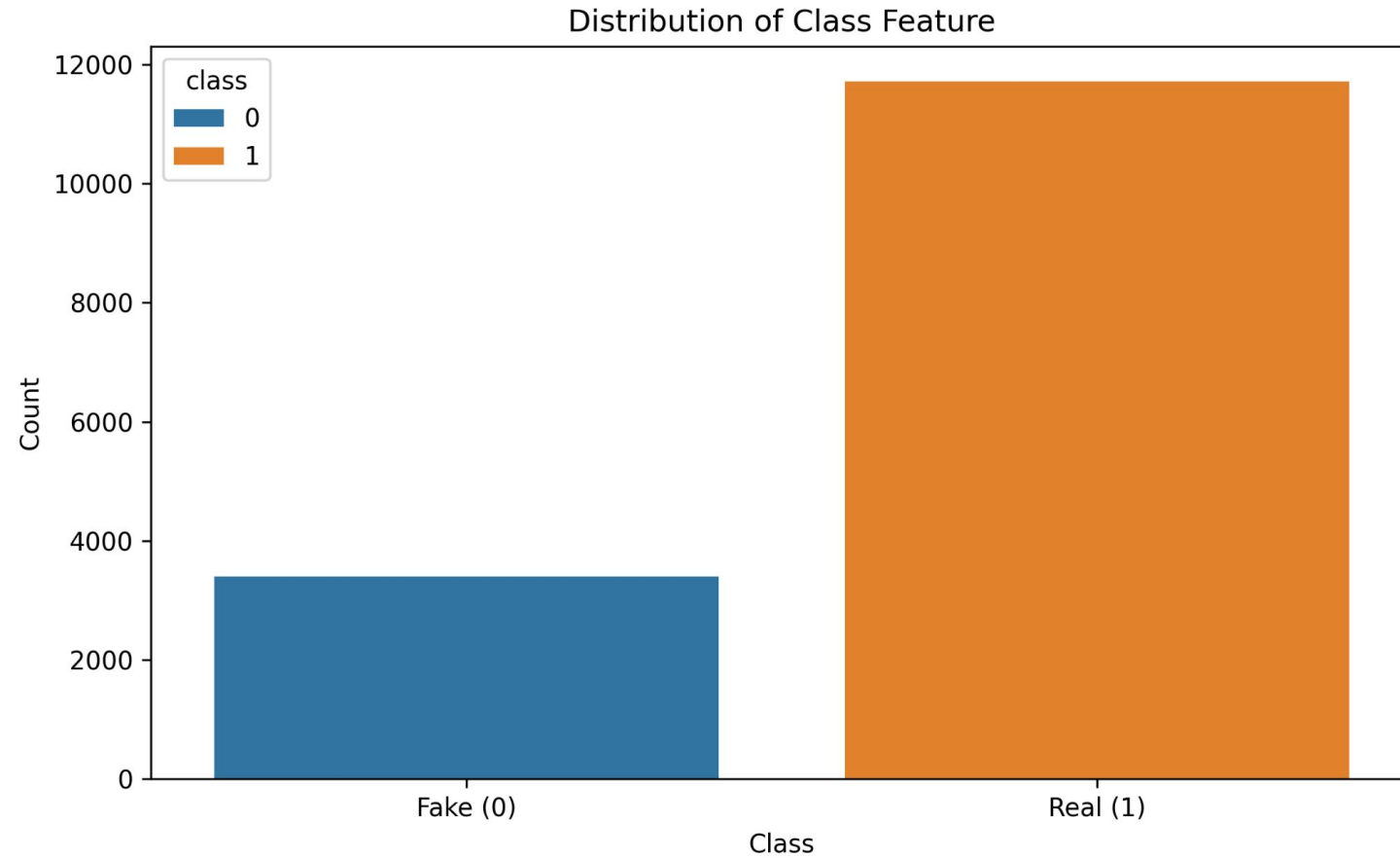
The 4 labelled datasets were cloned from;

<https://github.com/KaiDMML/FakeNewsNet/tree/master/dataset>

- The body text for news articles across the 4 datasets was scrapped based on respective `news\_url`.
- The enriched datasets were concatenated to yield a working dataset with 16,044 entities.

Key Challenge: Target variable class imbalance.

FINDINGS: EDA



The data is highly imbalanced:

- Fake articles = 22.5%
- Real articles = 75.5%

Strategies to address imbalance:

- Oversample minority class (SMOTE).
- Undersample majority class (RUS).

To determine the most impactful strategy; each ML model was trained using:

- 1) The imbalanced dataset.
- 2) SMOTE-balanced data.
- 3) RUS-balanced data.

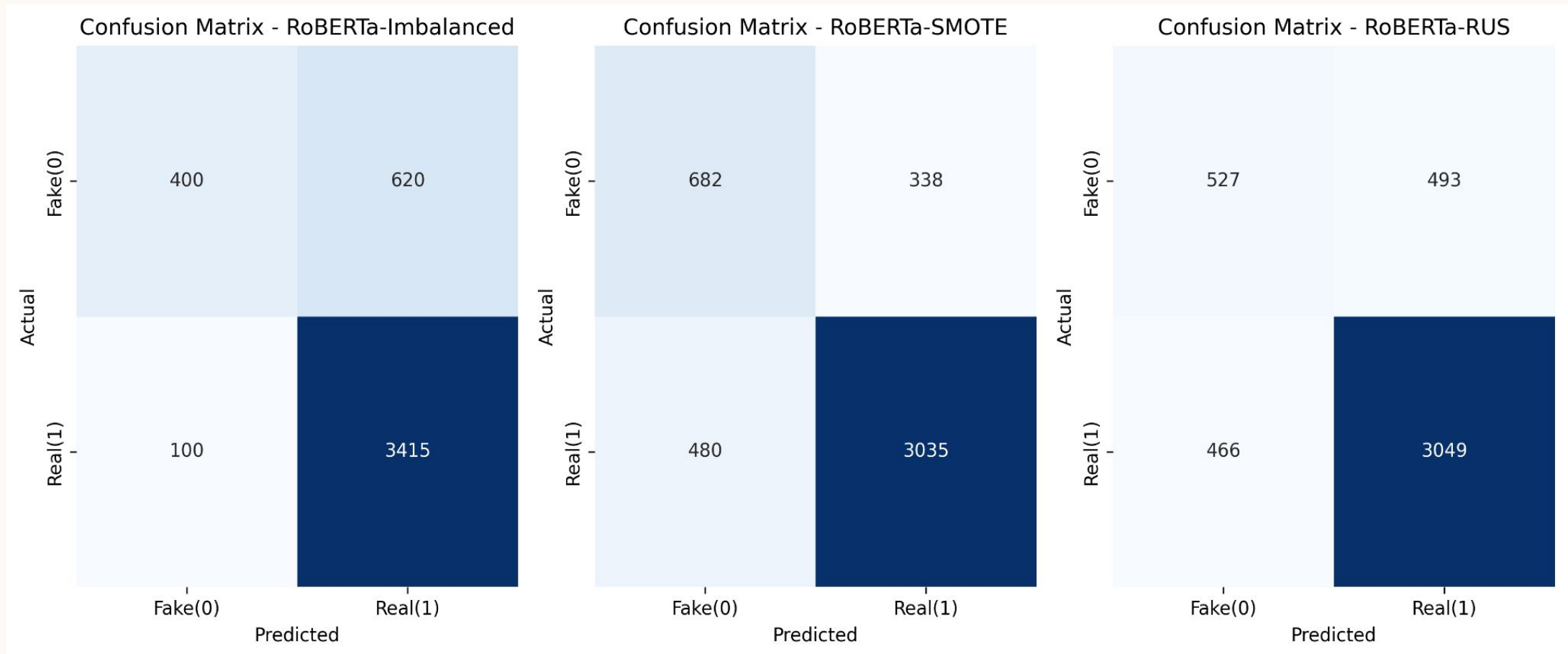
FINDINGS: MODELLING

Model	Best Configuration	Macro-averaged Recall	Recall (Fake)	Recall (Real)	Precision (Fake)	Precision (Real)	Accuracy	AUC
Logistic Regression	SMOTE-balanced data	0.7145	0.7029	0.7801	0.4812	0.9005	76.27%	0.815
XGBoost Ensemble model	RUS-balanced data	0.7427	0.7088	0.7767	0.4794	0.9019	76.14%	0.822
LSTM Neural Network	SMOTE-balanced data	0.7104	0.5784	0.8424	0.5157	0.8732	78.30%	0.792
Fine-tuned RoBERTa-base	SMOTE-balanced data	0.7660	0.6686	0.8634	0.5869	0.8998	81.96%	0.843

The fine-tuned RoBERTa-base transformer is the best-fit, and most generalizable model because it achieves:

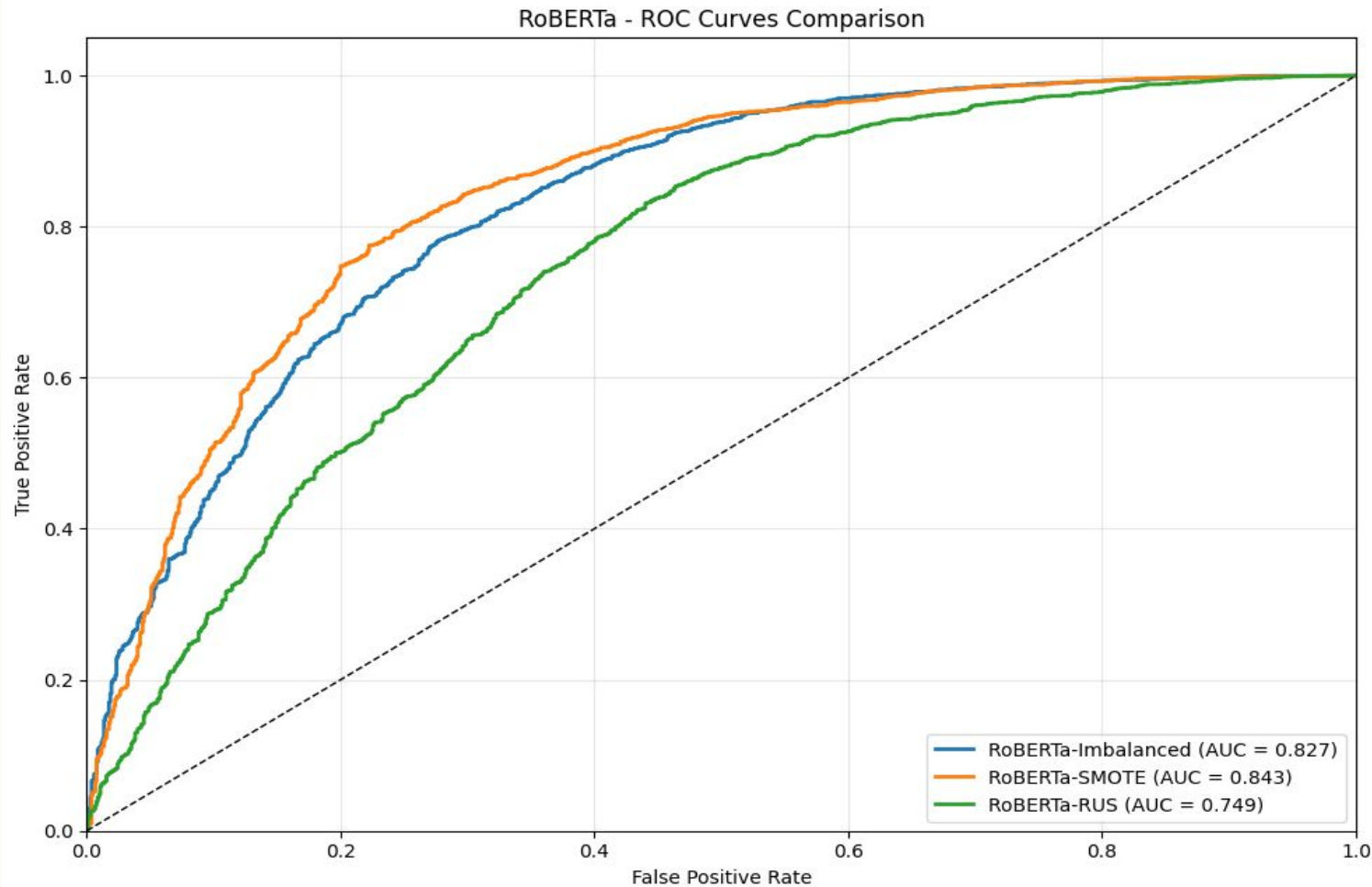
- ❖ **Highest Macro-averaged Recall:** Strikes an effective balance in catching both real and fake articles.
- ❖ **Highest Precision (Fake):** Highest confidence for “Fake Article” predictions.
- ❖ **Highest Accuracy:** Most accurate across either prediction.

FINDINGS: FINE-TUNED RoBERTa-base



→ The RoBERTa-base transformer fine-tuned with SMOTE-balanced data performs better on accurate predictions for “Fake News Articles”.

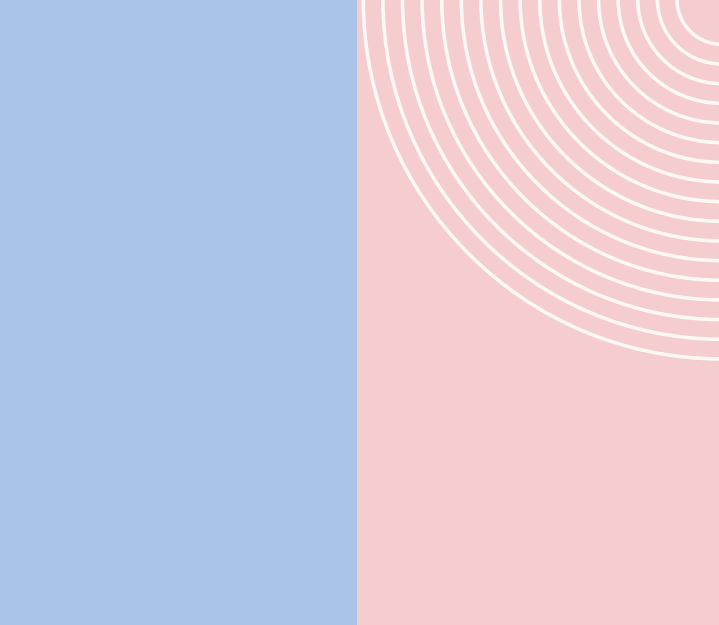
FINDINGS: FINE-TUNED RoBERTa-base



- ❑ The orange curve plots the True Positive rate against the False Positive rate for the RoBERTa transformer fine-tuned with SMOTE-balanced data.
- ❑ This curve hugs closest to the top-left corner affirming the model's superior ability to distinguish classes with minimal false positives.
- ❑ An AUC score of 0.843 indicates the model has an 84.3% chance to rank a randomly chosen positive instance higher than a randomly chosen negative instance.

CONCLUSIONS

- ❖ **Fake News Detection is possible:** We created a system that can effectively identify fake news articles.
- ❖ **Advanced Technology for better accuracy:** The system uses RoBERTa, which understands language to spot fake news accurately.
- ❖ **Balanced Approach:** The model balances fake and real news to improve detection without losing accuracy.
- ❖ **Continuous Improvement:** The system adapts over time to avoid overfitting and stay accurate.
- ❖ **Real-World Use:** The system can be used in apps or browser extensions for real-time fake news detection.
- ❖ **Expansion Potential:** The technology can be expanded to other languages and platforms.



RECOMMENDATIONS

- ❖ **Collaborate with Media Outlets:** Work with news organizations and social media platforms to incorporate this model into their systems for real-time fake news detection.
- ❖ **Educate the Public:** Raise awareness about fake news detection tools, and encourage people to use these technologies for more informed decision-making.
- ❖ **Expand into Multiple Languages:** Ensure the model can detect fake news in various languages to reach a broader audience globally.

NEXT STEPS

- ❖ Deploy the containerized fine-tuned RoBERTa model via AWS SageMaker for low-latency inference.
- ❖ Develop browser extensions and social media plugins using the model's API for real-time credibility scoring.
- ❖ Incorporate automation frameworks to flag low-confidence predictions for human review.





THANK YOU